Israfel Salazar

Mail: israfel.sr@gmail.com
Phone: (+45) 93938467
Website: israfelsr.github.io
LinkedIn: in/israfelsalazar
Scholar: Google Scholar

EDUCATION

UNIVERSITY OF COPENHAGEN
PhD in Multimodal Machine Learning
Advisor: Desmond Elliott
July 2024 - Present

ÉCOLE NORMALE SUPÉRIEURE (ENS)
MSc Applied Mathematics (MVA)
Oct 2022 - Oct 2023
Cum. Grade 16.26/20 (GPA 4.0)

UNIVERSITY PARIS-SACLAY
MSc Electrical Engineering, Machine Vision and AI
June 2020 - June 2022
Cum. Grade: 15.4/20 (GPA 3.7)
Best student of the major

UNIVERSITY OF CHILE
Mechanical Engineer (Engineering Diploma)
Completed June 2018
Cum. Grade: 6.2/7.0 (GPA 3.7)
6-year program

COURSEWORK

Probability and Statistics
Convex Optimization
Algorithms and Data Structures
Computational Statistics
Information Theory
3D Computer Vision
Deep Reinforcement Learning
Image and Video Processing
Natural Language Processing and Speech
CS231N CNNs for Visual Recognition

SKILLS

PROGRAMMING
Advanced:
Python • Pytorch • Pandas • Scikit-Learn
HuggingFace • Git • MuJoCo • LeRobot
Intermediate:
Tensorflow • SQL • Java • C++
Familiar:
R • ROS • CSS/HTML • VHDL • THREE.js

LANGUAGE
Spanish: Native
English: Fluent (C2)
French: Excellent command (C1)

EXPERIENCE

DXO LABS | Research Engineer Internship
April 2023 – Nov 2023 | Paris, France
- Developed a versatile codebase to create and evaluate different Diffusion architectures.
- Designed architecture and sampling technique, achieving state-of-the-art results in image deblurring and image sharpening.

LXMLS INSTITUTO SUPERIOR TÉCNICO | Core Monitor
Jun 2023 | Lisbon, Portugal
- Designed Transformer-Day exercises with cross-attention and multihead self-attention modules, providing easy-to-code and test implementations along with clear explanations.
- Provided guidance to students during coding lab sessions.

HUGGING FACE | ML Researcher & Collaborator
June 2022 – Dec 2022 | Paris, France
- Research project under the guidance of @Douwe Kiela and @Amanpreet Singh.
- Tested a unified approach to multimodal machine learning by processing text as images.
- Developed a pipeline for augmenting datasets modalities using generative models.
- Trained FLAVA to study performance impact of augmented multimodal datasets.

EXCELERATE | Software Engineering Intern
July 2021 – Sept 2021 | Copenhagen, Denmark
- Built a pipeline that fetched and summarized descriptions of university courses.
- Developed a scraper to obtain course descriptions and designed database to store them.
- Integrated a large language model to summarize the coursework experience of a student.

MUNDOS VIRTUALES | Robotics Engineer
April 2018 – Nov 2019 | Santiago, Chile
- Developed scripts for robotic arms involving decision-making and integration.
- Researched and manufactured state-of-the-art soft robotics for handling objects.
- Created a dataset to fine-tune a YOLO architecture for detecting ripe fruits.
- Simulated and modeled robotic arms using 3D environments for fruit harvesting.

RESEARCH PROJECTS

Pathways of Visual Information Flow in Vision-Language Models
NeurIPS 2026
- Identified two visual information pathways in VLMs, showing that routing is task- and data-dependent rather than fixed.
- Demonstrated that text mediation acts as a latent fallback pathway, recovering accuracy under image corruption even on tasks dominated by direct visual readout.

Long Story Short? Disentangling Compositionality and Long-Caption Understanding in Contrastive VLMs
Findings ACL 2026
- Curated an evaluation suite covering compositionality, long captions, and generalization.
- Conducted large-scale evaluation and fine-tuning of CLIP on long-caption datasets.
- Identified a bidirectional link between compositional and long-caption capabilities, improving retrieval performance.

Kaleidoscope: In-language Exams for Multilingual Vision Evaluation
ICLR 2026 | Collaboration with Cohere
- Built a multilingual VL-benchmark covering 10+ languages for in-language evaluation.
- Evaluated cross-lingual, multimodal, and culturally grounded reasoning in VLMs.

Specificity-Enhanced CLIP-Score for Long Image Caption Evaluation
EMNLP 2025 | Research Assistant @ MBZUAI
- Presented new evaluation metric for fine-grained visual-language alignment.
- Addressed CLIP's limitations, proposing a novel training objective for improved granularity.

A Quantitative Theory for Genomic Offset Statistics
MBE 2023 | Research Intern @ INRIA Statify
- Experimented with Bayesian Generative Deep Learning for genome viability prediction.
- Implemented Conditional Variational Autoencoders for genomic data generation.